

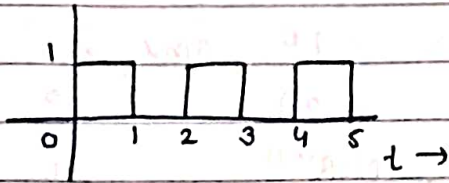


16/09/22

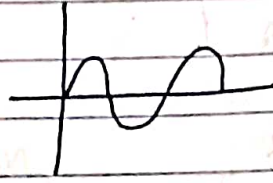
Discrete Mathematics

Books

Kenneth H. Rosen



Discrete Graph



Continuous/Analog Graph.

Propositional Calculus $\Rightarrow$

Proposition is a statement which can be either true or false, it is a declarative sentence.

Logic is the study of reasoning. Logic helps you to build better decision. It helps you to check the validity of program.

Symbols -

 N : Natural No. (set) R : Real No. (set) $P \cup Q$: Union, P or Q - $P+Q$ $P \cap Q$: Intersection, P and Q

$P \rightarrow Q$: If P is true and Q is also, P is antecedent and Q is consequent. If P then Q.

$P \leftrightarrow Q \equiv P \rightarrow Q \& Q \rightarrow P$: If P is true then Q and if Q is true then P. $P \equiv Q$

T : Tautology

F : Fallacy

 $\forall$: For all, $\forall n P(n)$ $\exists$: There exist

P	Q	$P \cup Q$	$P \cap Q$	P	Q	$P \cap Q$
0	0	0		0	0	0
0	1	1		0	1	0
1	0	1		1	0	0
1	1	1		1	1	1

$$P \oplus Q \equiv \bar{P}Q + P\bar{Q}$$

$$\equiv (P \cap Q) \cup (P \cap \neg Q)$$

10 And, No 1

x6R

NAND

NORON

$$\text{NOR} = \overline{P+Q} \equiv \neg(P \vee Q)$$

$$P \rightarrow 0 \Rightarrow PV0 \equiv (P + 0)$$

Any expression involving logical variables which are either true or false in case it is true it is named as tautology and if it false then fallacy or contradiction.

$>$: Not
 $\wedge$: And / Intersection
 $\vee$: Union / or
 $\rightarrow$ and $\leftrightarrow$

$$\begin{aligned} \text{Error} &= P \odot Q \equiv \bar{P} \bar{Q} + P Q \\ &\equiv (P \wedge Q) \vee (\bar{P} \wedge \bar{Q}) \end{aligned}$$

DELTA Notebook

Ques ① Show that $\neg(P \wedge Q) \vee Q$ is Tautology

Soln $\neg(P \wedge Q) \vee Q = \neg P \vee Q = Q \vee \neg P = Q \vee \neg Q \vee P = P \vee Q \vee \neg Q = P \vee 1 = 1$
 = Tautology

② Show that $P \wedge (Q \wedge \neg P)$ is Contradiction

③ $A \oplus B \oplus C$

④ $A \uparrow B \uparrow C$

⑤ Prove $X \oplus Y \equiv (X \wedge \neg Y) \vee (\neg X \wedge Y)$

P	Q	$\neg P$	$\neg Q$	$P \wedge Q$	$\neg(P \wedge Q)$	$\neg(P \wedge Q) \vee Q$
0	0	1	1	0	1	1
0	1	1	0	0	1	1
1	0	0	1	0	1	1
1	1	0	0	1	0	1

① Hence it is tautology

Q	$\neg P$	$P \wedge (Q \wedge \neg P)$
0	1	0
1	0	0
0	1	0
1	0	0

hence it is fallacy

④

A	B	C	$A \oplus B \oplus C$	$A \uparrow B \uparrow C$
0	0	0	0	1
0	0	1	1	1
0	1	0	1	1
0	1	1	0	1
1	0	0	1	1
1	0	1	0	1
1	1	0	0	1
1	1	1	1	0



⑤ $* x \oplus y = (x \wedge \neg y) \vee (\neg x \wedge y) = P$

⑥ Show $(P \oplus Q) \vee (P \wedge Q) \equiv (P \vee Q)$

⑦ Show $(P \vee Q) \oplus (P \wedge Q) \equiv (P \vee Q) \wedge (P \wedge Q)$

⑤	x	y	$x \oplus y$	$\neg x$	$\neg y$	$(x \wedge \neg y)$	$(\neg x \wedge y)$	P
	0	0	1	1	1	0	0	0
	0	1	0	1	0	0	1	1
	1	0	0	0	1	1	0	1
	1	1	0	0	0	0	0	0

③	$A \oplus B \oplus C$
	0
	1
	1
	0
	1
	0
	0
	1

21/09/22

Logical Equivalences & Interferences

		disjunction	conjunction
P	$\supset P$	$P \vee \supset P$	$P \wedge \supset P$
T	F	T	F
F	T	T	F

Tautology
Fallacy

Q4. $\supset (P \vee Q) \equiv \supset P \wedge \supset Q$

$\Rightarrow$	$\overline{P+Q}$	$\&$	$\overline{P} \cdot \overline{Q}$	$\overline{P+Q} \equiv \overline{P} \cdot \overline{Q}$		
$P \vee Q$	Q	$P \vee Q$	$\supset (P \vee Q)$	$\supset P$	$\supset Q$	$\supset P \wedge \supset Q$
T	F	T	F	F	T	F
F	T	T	F	T	F	F

$$\supset (P \vee Q) \equiv \supset P \wedge \supset Q$$

— x — x — x — x — x — x — x — x — x — x — x —

(1) $P \wedge T \equiv P$
 $P \vee F \equiv P$ } Identity Law

(7) $P \vee (Q \wedge R) = (P \vee Q) \wedge (P \vee R)$
 $P \wedge (Q \vee R) = (P \wedge Q) \vee (P \wedge R)$ } Distributive Law

(2) $P \vee T = T$
 $P \wedge F = F$ } Domination Law

(8) De-Morgan's Law
 8.1) $\supset (P \wedge Q) \equiv \supset P \vee \supset Q$
 8.2) $\supset (P \vee Q) \equiv \supset P \wedge \supset Q$

(3) $P \vee P = P$
 $P \wedge P = P$ } Idempotent Law

8.2) $P \rightarrow Q \equiv \supset P \vee Q$

(4) $\supset (\supset P) = P$ } Double Negation

(5) $P \vee Q = Q \vee P$
 $P \wedge Q = Q \wedge P$ } commutative Law

(6) $P \vee (Q \vee R) = (P \vee Q) \vee R$
 $P \wedge (Q \wedge R) = (P \wedge Q) \wedge R$ } Associative Law

$$(p \vee q) \wedge (\neg p \vee \neg q) \Rightarrow (p \rightarrow \neg q) \wedge (\neg p \rightarrow q) \Rightarrow (p \vee q) \wedge (\neg p \vee \neg q)$$

$$p \vee p \cdot q$$

Date _____

Page No. _____



$$\begin{aligned} (9) \quad p \vee (p \wedge q) &= p \\ p \wedge (p \vee q) &= p \end{aligned} \quad \left. \begin{array}{l} \text{Absorption} \\ p \cdot (p + q) = p \cdot p + p \cdot q = p(1 + q) = p \end{array} \right\} \Rightarrow p \vee (p \wedge q) = p(1 + q) = p(1) = p$$

$$(10) \quad \begin{aligned} p \vee \neg p &= T \\ p \wedge \neg p &= F \end{aligned} \quad \left. \begin{array}{l} \text{Negation} \\ \text{Law} \end{array} \right\}$$

$$(11) \quad p \rightarrow q \equiv \neg p \vee q \quad (\text{Second Law of De-Morgan's})$$

$$(12) \quad p \rightarrow q \equiv \neg q \rightarrow \neg p \quad (\text{Contrapositive Law})$$

$$(12) \quad \begin{aligned} p \vee q &\equiv \neg p \rightarrow q \\ p \wedge q &\equiv \neg(p \rightarrow \neg q) \end{aligned}$$

Based on

De-Morgan's Law.

$$(13) \quad \neg(p \rightarrow q) \equiv p \wedge (\neg q)$$

$$(14) \quad \begin{aligned} (p \rightarrow q) \wedge (p \rightarrow r) &\equiv p \rightarrow (q \wedge r) \\ (p \rightarrow r) \wedge (q \rightarrow r) &\equiv (p \vee q) \rightarrow r \\ (p \rightarrow q) \vee (p \rightarrow r) &\equiv p \rightarrow (q \vee r) \\ (p \rightarrow r) \vee (q \rightarrow r) &\equiv (p \wedge q) \rightarrow r \end{aligned}$$

$$\Rightarrow p \rightarrow (q \wedge r) = \neg p \vee (q \wedge r) = (\neg p \vee q) \wedge (\neg p \vee r) = \overline{p + (q \cdot r)}$$

$$\begin{aligned} (14) \quad (\bar{p} + q)(\bar{p} + r) &= \bar{p} + \bar{p}r + q\bar{p} + qr \\ &= \bar{p}(1 + r) + q(\bar{p} + r) = \bar{p}(1) + q(\bar{p} + r) \\ &= \bar{p} + q\bar{p} + qr = \bar{p}(1 + q) + qr = \bar{p} + qr \end{aligned}$$

Q. show that $\neg(p \vee (\neg p \wedge q))$ & $\neg p \wedge \neg q$ are logically equivalent

Sol. $\neg p \wedge \neg q$

$$\begin{aligned} \neg(p \vee (\neg p \wedge q)) &= \neg((p \vee \neg p) \wedge (p \vee q)) \\ &= \neg[1 \wedge (p \vee q)] = \neg[p \vee q] = (\neg p \wedge \neg q) \end{aligned}$$



Sol^y $\overline{p + \bar{p}q} \equiv (\bar{p}) \cdot (\overline{\bar{p}q})$
 $\equiv (\bar{p}) \cdot (\bar{\bar{p} + q}) = (\bar{p}) \cdot (p + \bar{q})$
 $\equiv \bar{p} \cdot p + \bar{p} \cdot \bar{q} = \bar{p} \cdot \bar{q} = \bar{p} \wedge \bar{q}$

Q^y - Show that

$(p \wedge q) \rightarrow (p \vee q)$ is a tautology.

Sol^y $\neg (p \wedge q) \vee (p \vee q)$
 $\Rightarrow (\neg p \vee \neg q) \vee (p \vee q) \rightarrow \text{De-Morgan's}$

$\Rightarrow (\neg p \vee p) \vee (\neg q \vee q)$

$\Rightarrow (\neg p \vee p) \vee (\neg q \vee q) \rightarrow \text{Negation}$

$\Rightarrow T \vee T = T$

Hence proved it is tautology.

22/09/22

Predicates

TANNU / is an honest / student
 subject predicate domain
 X H(n)

Propositional logic can't adequately explain the meaning of the statement PL given an expression to define of wide variety of statement in Maths & CS, it also helps us to explore relationship b/w objects

eg -

$y = x + 3$

$y = n^2$

$y > n^2 \Rightarrow y$ is greater than square of n

subject predicate



Q- Computer x is under attack by an intruder.

Q. Let $P(x)$ denote statement $x > 3$ what is the truth value of $P(4)$ & $P(2)$

Sol	$P(x)$	$x > 3$	
	$P(4)$	$4 > 3$	True
	$P(2)$	$2 > 3$	False

Q. Let $A(n)$ denote Computer x is under attack by an intruder. Suppose that the computers on campus are only CS2 and MATH are currently under attack by intruders, what is the truth value of $A(CS1)$

Sol. We obtain the soln by setting the value as CS1. Since CS1 is not in the list of computer attack so $A(CS1)$ will be false & $A(CS2)$ & $A(MATH)$ will be true

Q- Let $Q(x, y)$ $x = y + 3$ what are the truth value of $Q(1, 2)$ & $Q(3, 0)$

Sol	$Q(1, 2)$	$1 = 5$	False
	$Q(3, 0)$	$3 = 3$	True

Quantification

$\forall x$	$\exists x$	$\forall x \rightarrow$ Universal Quantification
For all	There exist	$\exists x \rightarrow$ Existential "
All, every, each	some, few, one	

$$x = \{x_1, x_2, x_3, \dots\}$$

$$P(x) \begin{cases} T \\ F \end{cases}$$

$x \rightarrow$ Domain



$$\left[\begin{array}{l} \forall n \ P(n) \equiv P(1) \wedge P(2) \wedge P(3) \wedge P(4) \dots P(n) \\ \exists n \ P(n) \equiv P(1) \vee P(2) \vee P(3) \vee P(4) \dots P(n) \end{array} \right]$$

$\forall n P(n) \rightarrow P(n)$ is true for every x .

$\exists n P(n) \rightarrow$ There exist an n for which $P(x)$ is true.

Q Let $Q(n)$ be the statement $n < 2$, what is the truth value for all n , $Q(n)$, where domain is all real no.

Soln

$\forall n P(n)$	$\begin{matrix} T^x \\ F \end{matrix}$	$P(0)$, For $n=0$, $0 < 2$ True
		$P(1)$, For $n=1$, $1 < 2$ T
		$P(2)$, For $n=2$, $2 < 2$ F
		$\vdots$

Counting

$\Rightarrow$ False

Q Let $P(n)$ denote the statement $n > 3$ what is the truth value $\exists n P(n)$, where domain is all real no.

Soln

$P(0)$	$n=0$	$0 > 3$	F
$P(1)$	$n=1$	$1 > 3$	F
$P(2)$	$n=2$	$2 > 3$	F
$P(3)$	$n=3$	$3 > 3$	F
$P(4)$	$n=4$	$4 > 3$	T
$P(5)$	$n=5$	$5 > 3$	T

$\Rightarrow$ True

Negation of Quantifiers-

$$\neg \forall n P(n) \equiv \exists n \neg P(x)$$

Not all x , $P(n) \equiv$ There exist some n for which $P(x)$ is false.



Q. What are the negation of the statements

① There is an honest politician.

② All American eats cheese burger

Sol. $\neg \exists n P(n) \equiv \forall n \neg P(n)$

① $\neg \exists n H(n) \equiv \forall n \neg H(n)$

There doesn't exist For all politician honesty doesn't exist
an honest politician

② $\neg \forall n C(n) \equiv \exists n \neg C(n)$

Not all American There exist American, who doesn't
eats cheese burger eats cheese burger.

27/09/22

Rules of Inference

I) Modus Ponens $\begin{array}{l} P \\ P \rightarrow q \\ \hline \therefore q \end{array}$	Transitive OR II Hypothetical Syllogism $\begin{array}{l} P \rightarrow q \\ q \rightarrow r \\ \hline \therefore P \rightarrow r \end{array}$
II Modus Tollens $\begin{array}{l} \neg q \\ P \rightarrow q \\ \hline \therefore \neg P \end{array}$	III Disjunctive syllogism $\begin{array}{l} P \vee q \\ \neg P \\ \hline \therefore q \end{array}$
III Addition $\begin{array}{l} P \\ \hline \therefore P \vee q \end{array}$	VI Disjunction or syllogism $\begin{array}{l} P \wedge q \\ \hline \therefore P \text{ or } P \wedge q \rightarrow P \\ \text{Simplification} \end{array}$



VII Simplification

$$\frac{p \vee q}{p}$$

$$\therefore p$$

VIII conjunction

$$p$$

$$q$$

$$\therefore p \wedge q$$

IX Resolution

$$p \vee q$$

$$\neg p \vee r$$

$$\therefore q \vee r$$

Q. State which Rule of Inference is the basis of following argument

① It is below freezing now therefore it is either below freezing or raining now. Addition

② If it rains today, then we'll not have a barbeque today. If we don't have a barbeque today then we'll have barbeque tomorrow, therefore if it rains today we will have barbeque tomorrow.
Hypothetical Syllogism

③ It is not sunny this afternoon and it is colder than yesterday.

- we'll go swimming only if it is sunny

- If we do not go swimming, then we take a canoe trip

- If we take a canoe trip, we'll be home by sunset.

- We'll be home by sunset.

$$\rightarrow \bar{p} \wedge q \Rightarrow \neg p \text{ - simplification}$$

$$x \rightarrow p \Rightarrow \neg x \text{ - Modus tollens}$$

$$\bar{x} \rightarrow y \Rightarrow y \text{ - Modus ponens}$$

$$x \rightarrow t \Rightarrow t \text{ - Modus ponens}$$



- Q. Show that the premises, if you send me an email msg, then I'll finish writing a program.
- If you don't send an email msg, then I'll go to sleep early
 - If I go to sleep early, then I'll wake up feeling fresh.

Concl - If I don't finish writing the program then I'll wake up feeling refreshed

$$\Rightarrow \begin{array}{l} p \rightarrow q \\ \bar{p} \rightarrow r \\ r \rightarrow s \end{array} \equiv \begin{array}{l} \Rightarrow q \rightarrow \Rightarrow p \\ \therefore \\ \end{array} \quad \left| \begin{array}{l} p \rightarrow q \\ \Rightarrow q \rightarrow \Rightarrow p \\ \text{Contrapositive.} \end{array} \right.$$

$$\Rightarrow \left. \begin{array}{l} \Rightarrow q \rightarrow \Rightarrow p \\ \Rightarrow p \rightarrow r \end{array} \right\} \text{from Hypothetical } \Rightarrow \Rightarrow q \rightarrow r$$

syllogism

$$\Rightarrow \left. \begin{array}{l} \Rightarrow q \rightarrow r \\ r \rightarrow s \end{array} \right\} \text{from H.S } \Rightarrow \Rightarrow q \rightarrow s$$

$\begin{array}{l} \# \\ p \rightarrow q \equiv \Rightarrow q \rightarrow \Rightarrow p \\ \Rightarrow p \rightarrow \Rightarrow q \\ q \rightarrow p \end{array}$	Contrapositive Inverse Converse
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28/09/22

Q-1. If it is Wednesday the smart mart will be crowded

For clusion It is wed, thus the smart mart is crowded.

$$\begin{array}{l} p \rightarrow c \\ p \rightarrow a \\ \hline q \rightarrow c \end{array}$$

- Q1. If it rains today the local office will be closed.
2. The local office is not closed today, it doesn't rain today.

$$\frac{p \rightarrow q}{\neg q \rightarrow \neg p}$$

Q- Bhavika will work in enterprise this summer. Therefore this summer Bhavika will work in enterprise she will go to the beach.

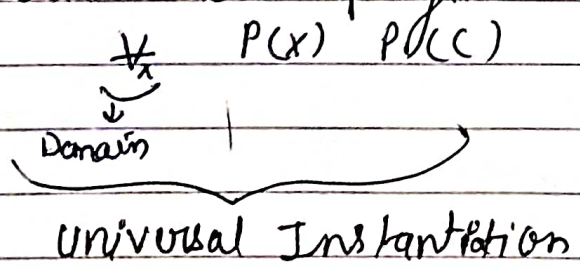
- Addition

- Q- If I go for a balance diet then I will be fit.
- Q- If I'll be fit then I'll remain healthy.
- (C) Therefore I'll go for balance diet, I'll remain healthy.

* Hypothetical syllogism

Q- All students in the science class have taken a course in physics and Mary is a student in a class.

Implies the conclusion that Mary has a course in physics.



Instance
C: Mary



Rules of Inference for Quantifiers.

① Universal Instantiation

$$\frac{\forall x P(x)}{\therefore P(c)} \leftarrow \text{Instantiation step}$$

①. It is either colder than Himalaya today or the pollution is harmful.

②. It is hotter than Himalaya today, therefore the pollution is harmful.

$$\begin{array}{l} \text{Let } p \vee q \\ \underline{p} \quad \text{D.S} \\ \hline q \end{array}$$

$$\text{Qn. } \frac{(p \wedge q) \vee r}{r \rightarrow s}$$

$$(pq \vee r)(\bar{r} + s) = pq\bar{r} + pq s + r\bar{r} + rs$$

$$= pq\bar{r} + pq s + rs = p(\cancel{q\bar{r}} + s)$$

$$= pq(\bar{r} + s) + rs$$

Apply Distributive law

$$\frac{(p \vee r) \wedge (q \vee r)}{(p \vee r) \wedge (q \vee r)}$$

$$\therefore \underline{p \vee s}$$

Q. Jai is an awesome student. Jai is also a good dancer. Therefore Jai is an awesome student and a good dancer.

Sol. Conjunction

$$\frac{p}{q}$$

$$\frac{p}{p \wedge q}$$

A parcel is out for a trip or it is not snowing and it is snowing or Raju is playing chess. What this implies.

Ru



29/9/22

Rules of Inference for Quantified Expressions

① Universal Instantiation

$$\frac{\forall x P(x)}{\therefore P(c)}$$

where x is the domain
 c is the instance/object
 $P(x)$ is the premise
 $\forall n$ for all x elements.

② Universal Generalization

$$\frac{P(c)}{\therefore \forall x P(x)}$$

③ Existential Instantiation

$$\frac{\exists x P(x)}{\therefore P(c)}$$

④ Existential Generalization

$$\frac{P(c)}{\therefore \exists x P(x)}$$

Q4-1 Show that the premise everyone in this D-M class ^{has} taken a course in CS and Maria is a student in the class. Implies the conclusion Maria has taken the course in CS

$$\Rightarrow \begin{array}{l} \forall x (D(x) \rightarrow C(x)) \quad \text{--- ①} \\ D(\text{Maria}) \quad \text{--- ②} \\ \hline D(\text{Maria}) \rightarrow C(\text{Maria}) \quad \text{--- ③} \end{array}$$

$\therefore C(\text{Maria})$ (Using Modus ponens in ③ & ②)



- Q. Show that the premise. A student in this class has not read the ~~text~~ book.
- Everyone in this class passed the first class exam.
- (C) - Someone who passed the class has not read the book.

$$\Rightarrow \begin{cases} \exists x (C(x) \wedge \neg B(x)) & \text{--- (1)} \\ \forall x (C(x) \rightarrow P(x)) & \text{--- (2)} \end{cases}$$

To Prove

$$\begin{cases} \exists x (P(x) \wedge \neg B(x)) & \text{--- (3) using (E.I)} \\ C(a) \wedge \neg B(a) & \text{--- (4) using (U.I)} \\ C(a) \rightarrow P(a) & \text{--- (5)} \end{cases}$$

$C(a)$ --- (4) Simplification (eqn-3)

Modus Ponens on (4) & (5), we get

$P(a)$ --- (A)

Applying simplification on eqn (3) --- (B)

$\neg B(a)$ --- (B)

Apply 1st Conjunction then Existential generalization

Apply Conj. - $P(a) \wedge \neg B(a)$

Apply EG = $\exists x (P(x) \wedge \neg B(x))$

Q. Mary loves everyone

$$\forall x \text{ loves } (Mary, x)$$

$$\forall x (\text{person } x) \rightarrow \text{loves } (Mary, x)$$

2- No one talks.

$$\neg \exists x \text{ talks } (x) \quad \text{or} \quad \forall x \neg \text{talks } (x)$$

3- Everyone loves himself

$$\forall x \text{ loves } (x, x)$$

- (4) Everyone loves everyone
 $\forall x \forall y \text{ loves}(x, y)$
- (5) Everyone loves everyone except himself
 $\forall x \forall y (x \neq y \rightarrow \text{loves}(x, y))$
 $\forall x \forall y (x = y \rightarrow \neg \text{loves}(x, y))$
- (6) Every student smiles
 $\forall x (\text{stu}(x) \rightarrow \text{smiles}(x))$
- (7) Every student except George smile.
 $\forall x (\text{stu}(x) \wedge x \neq \text{George} \rightarrow \text{smiles}(x))$
- (8) Everyone walks or talks
 $\forall x (\text{walks}(x) \vee \text{talks}(x))$
- (9) Every student walks or talks
 $\forall x (\text{stu}(x) \rightarrow (\text{walks}(x) \vee \text{talks}(x)))$
- (10) Every student who walks talks
 $\forall x (\text{stu}(x) \wedge \text{walks}(x) \rightarrow \text{talks}(x))$